

Problem 1

Assume A is a Noetherian ring.

Let I be an ideal in $A[[X]]$.

Consider $I_n = \{a \in A \mid \exists f(X) \in I \text{ s.t. } f(X) = aX^n + a_{n+1}X^{n+1} + 1 + \dots\}$

Let $a, b \in I_n$ then $aX^n + \dots, bX^n + \dots \in I$ thus $(a + b)X^n + \dots \in I$ thus $(a + b) \in I_n$

Let $a \in I_n$ and $r \in A$ then $aX^n + \dots \in I$ thus $raX^n + \dots \in I$ thus $ra \in I_n$

Therefore, I_n is an ideal of A thus it is finitely generated.

Consider I_n and I_{n+1} . If $a \in I_n$ then $aX^n + \dots \in I$ thus when you multiply by X you get $aX^{n+1} + \dots \in I$ thus $a \in I_{n+1}$.

Therefore, $I_n \subseteq I_{n+1}$ thus we get an ascending chain of ideals. Since A is Noetherian it has to stabilize at some point so let I_N be the ideal in A where it stabilizes (the lowest number). Thus, I_N is finitely generated by $a_1, \dots, a_n$

Now pick $f_i(X) = a_iX^n + \dots$

Consider $(f_1, \dots, f_n)$

Now, take any $f(X) \in I$

If the lowest nonzero term of $f(X)$ is at a degree less than N , then since there are only finitely many such degrees $(0, 1, \dots, N-1)$, we can generate all these parts by finitely many elements of I coming from the ideals $I_0, I_1, \dots, I_{N-1}$.

If the lowest nonzero term of $f(X)$ is bX^k with $k \geq N$, then $b \in I_k = I_N$ (because the chain $I_0 \subseteq I_1 \subseteq I_2 \subseteq \dots$ stabilizes at I_N). Since I_N is finitely generated by $a_1, \dots, a_n$, we have

$$b = c_1a_1 + \dots + c_na_n.$$

For each a_i there is a corresponding $f_i(X) = a_iX^N + \dots \in I$. Multiplying $f_i(X)$ by X^{k-N} yields a series with lowest nonzero term a_iX^k , and then

$$g(X) = c_1X^{k-N}f_1(X) + \dots + c_nX^{k-N}f_n(X)$$

has lowest nonzero term bX^k matching that of $f(X)$. Subtracting $g(X)$ from $f(X)$ cancels the lowest nonzero term, and by repeating this process (which terminates in finitely many steps), we express $f(X)$ as an $A[[X]]$ -linear combination of finitely many generators. Thus, every $f(X) \in I$ is generated by a finite set, proving that I is finitely generated and that $A[[X]]$ is Noetherian.

Problem 2